

Element AD

$\Pi_e(u_e)$

exact local gradient and Hessian
higher-order elements can be costly

Constitutive AD

$\psi(\varepsilon_q)$

quadrature-point tangent assembly
exact local constitutive derivatives

Colored SFD

rank-local probe HVPs only where needed
parallel coloring / recovery
useful when exact Hessians are too expensive